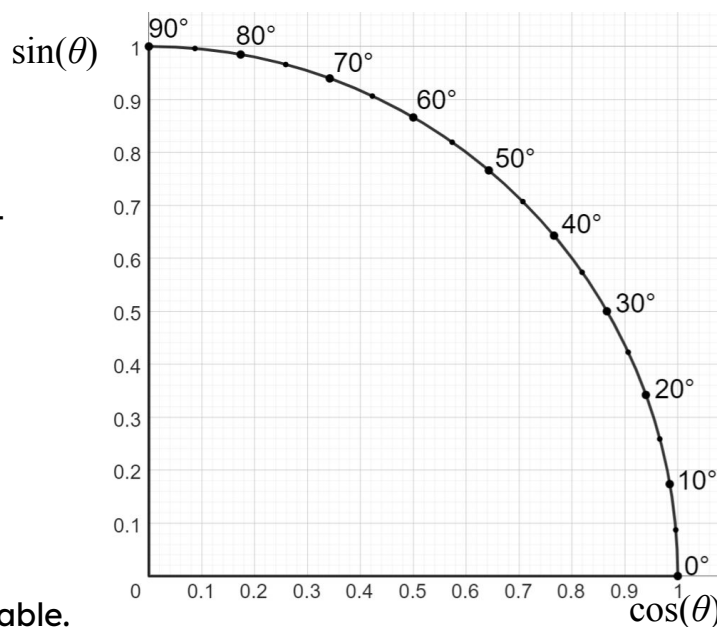




The unit circle

Task A: Introduction to sine and cosine

The diagram shows the sector of the unit circle that lies in the first quadrant.



1) a) Use the diagram to complete the table.

θ°	0	10	20	30	40	50	60	70	80	90
$\sin(\theta)$										

b) What happens to the value of $\sin(\theta)$ as the angle increases in this quadrant?

c) Use your table to find the angle where $\sin(\theta) = 0.5$.

d) Use the diagram to find the angle where $\sin(\theta) = 0.8$.

2) a) Use the diagram to complete the table.

θ°	0	10	20	30	40	50	60	70	80	90
$\cos(\theta)$										

b) What happens to the value of $\cos(\theta)$ as the angle increases in this quadrant?

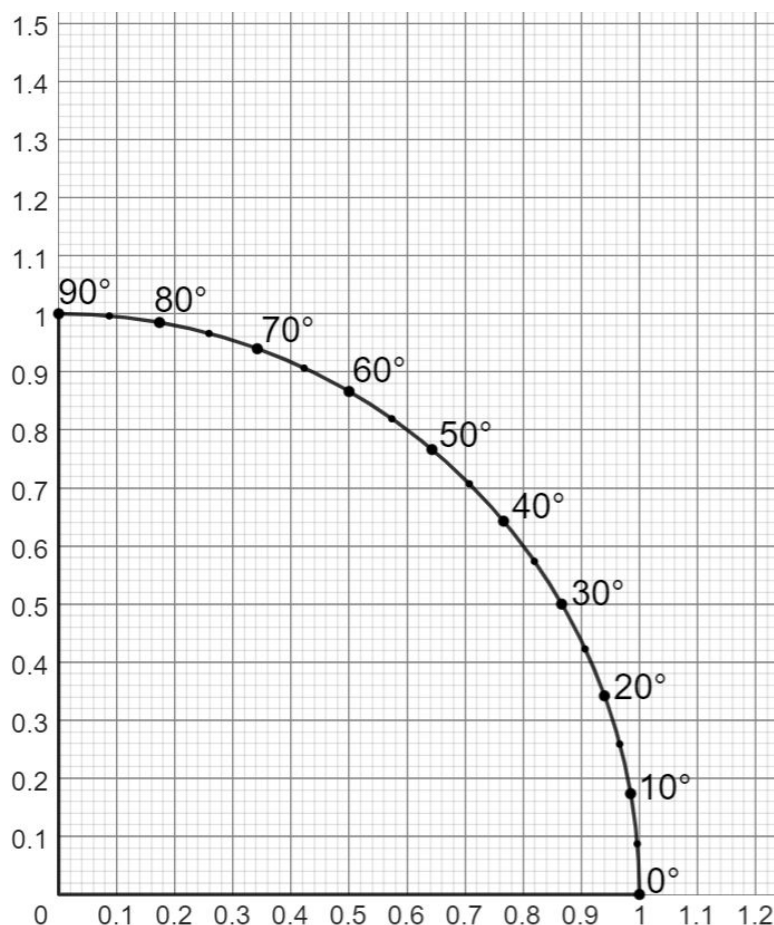
c) Find the angle where $\cos(\theta) = 0.5$.

d) Find the value of θ where $\cos(\theta) = \sin(40^\circ)$.

e) Find the angle where $\cos(\theta) = \sin(\theta)$.



Task B: Introduction to tangent



1) The diagram shows the first quadrant of the unit circle.

a) Draw a tangent at $x = 1$.

b) Find the value of $\tan(35^\circ)$

c) Find the value of $\tan(55^\circ)$

d) Use the diagram to find the angle where $\tan(\theta) = 1$.

e) Use the diagram to find the angle where $\tan(\theta) = 0.5$.

For an interactive version of the diagram, click the link: [unit circle in first quadrant](#)

2) a) Complete the table.

θ°	30	40	50
$\sin(\theta)$			
$\cos(\theta)$			
$\tan(\theta)$			

b) For each angle, divide the value of $\sin(\theta)$ by the value of $\cos(\theta)$. What do you notice?